This warping, in turn, affects other objects moving in the vicinity of the sun, as they now must traverse the distorted spatial fabric. Using the rubber membrane-bowling ball analogy, if we place a small ball-bearing on the membrane and set it off with some initial velocity, the path it will follow depends on whether or not the bowling ball is sitting in the center. If the bowling ball is absent, the rubber membrane will be flat and the ball bearing will travel along a straight line. If the bowling ball is present and thereby warps the membrane, the ball bearing will travel along a curved path. In fact, ignoring friction, if we set the ball bearing moving with just the right speed in just the right direction, it will continue to move in a recurring curved path around the bowling ball—in effect, it will "go into orbit." Our language presages the application of this analogy to gravity.

The sun, like the bowling ball, warps the fabric of space surrounding it, and the earth's motion, like that of the ball bearing, is determined by the shape of the warp. The earth, like the ball bearing, will move in orbit around the sun if its speed and orientation have suitable values. This effect on the motion of the earth is what we normally would refer to as the gravitational influence of the sun, and is illustrated in Figure 3.5. The difference, now, is that unlike Newton, Einstein has specified the mechanism by which gravity is transmitted: the warping of space. In Einstein's view, the gravitational tether holding the earth in orbit is not some mysterious instantaneous action of the sun; rather, it is the warping of the spatial fabric caused by the sun's presence.

This picture allows us to understand the two essential features of gravity in a new way. First, the more massive the bowling ball, the greater the distortion it causes in the rubber membrane; similarly, in Einstein's description of gravity the more massive an object is, the greater the distortion it causes in the surrounding space. This implies that the more massive an object, the greater the gravitational influence it can exert on other bodies, precisely in accord with our experiences. Second, just as the distortion of the rubber membrane due to the bowling ball gets smaller as one gets farther from it, the amount of spatial warping due to a massive body such as the sun decreases as one's distance from it increases. This, again, jibes with our